HYPERTRADE WHITEPAPER

# A Non-Custodial Interface for Global Trading and Stablecoin Finance

HyperTrade is building a mobile-first interface that makes advanced onchain markets usable by everyday users while preserving self-custody, transparent execution, and user-controlled authorization.

| MARKET | CUSTODY | RAILS | TRADING | FINANCE |
|---|---|---|---|---|
| **Global** | **Non-custodial UI** | **USDC + IBAN** | **Hyperliquid** | **UR.APP** |

**Vision: make onchain trading and stablecoin finance feel as simple as a modern fintech app, while keeping the user in control of funds and signatures.**

Draft generated 2026-07-13 • Prepared for investors, partners, and early users

# Contents

# 1. Vision and Goals

HyperTrade is designed for a global audience. The target user is not only the crypto-native trader who understands perpetual futures, RPCs, chain IDs, and gas. It is also the mobile user who wants dollar-denominated balances, intuitive market access, clear risk controls, and eventually the ability to use stablecoins for daily financial life.

The long-term goal is to make HyperTrade a casual, easy-to-use trading and stablecoin finance interface for all levels of users: beginners who need guided education, active traders who need speed, and professionals who want mobile access to onchain markets without giving up custody.

- **For entry users:** remove wallet friction, explain risk, provide demo practice, and make deposits feel like a modern fintech flow.
- **For active traders:** provide fast order entry, portfolio monitoring, one-tap trading setup, and direct Hyperliquid execution.
- **For global stablecoin users:** evolve from trading balances into cards, payments, and bill-pay style everyday utility.
- **For partners and investors:** build at the intersection of onchain execution, stablecoin adoption, mobile distribution, and AI-assisted finance.

> **The product goal is not only to trade more assets. It is to make onchain financial infrastructure usable globally without turning the app into a custodian.**

# 2. Why Now: Global Stablecoins and Onchain Markets

Stablecoins have become one of crypto's clearest product-market-fit categories: dollar-like balances, global transferability, and fast settlement. At the same time, onchain market infrastructure is moving beyond crypto-only speculation into equities, commodities, FX, and index exposure.

HyperTrade sits at the edge of this convergence. A user who starts with a USDC trading balance can later become a stablecoin payment user. A user who starts with a simple market order can later use advanced perps, portfolio tools, AI analysis, or a card product.

This matters globally. In many markets, access to stable dollar balances, always-on markets, and cross-border payments is more useful than another zero-utility token. HyperTrade's thesis is that useful financial rails will outperform purely narrative-driven crypto applications.

| STABLECOINS | PERPS | MOBILE | AI |
| --- | --- | --- | --- |
| **payments + savings** | **global market access** | **mass distribution** | **decision support** |

# 3. Product: A Trading Interface for Every Level

HyperTrade is an interface, not a traditional broker. The app helps users prepare, sign, and submit transactions while keeping funds in user-controlled wallets and routing execution to external onchain infrastructure.

| User Need | HyperTrade Response |
|---|---|
| Easy onboarding | Privy email, Google, and Apple login with automatic embedded wallet creation. |
| Simple balances | Wallet balance, trading balance, and portfolio screens with clear transfer states. |
| Fast trading | Hyperliquid API-wallet setup enables one-tap order signing after user approval. |
| Risk awareness | Margin previews, leverage controls, liquidation estimates, reduce-only guards, and education-first copy. |
| Future everyday finance | IBAN accounts, FX, and debit cards via UR.APP (External Wallet Access mode). |

## Experience principles

- **Start simple, reveal depth:** new users can place a straightforward trade, while advanced users still get leverage, margin mode, order editing, and portfolio controls.
- **Make balances legible:** wallet funds, trading funds, and future card balances should never be ambiguous.
- **Respect user control:** every sensitive action remains user-authorized, even when the app hides operational complexity.

# 4. Architecture Overview

The current architecture is intentionally modular. User identity and wallet signing live in the mobile app. Backend services coordinate relays, market data, notifications, and operational state. Hyperliquid is the active execution venue. Arbitrum USDC is the stablecoin settlement rail. IBAN-backed fiat accounts and card rails are delivered through **UR.APP** ([docs.ur.app](https://docs.ur.app)) via External Wallet Access — user-signed flows with regulated fiat-token rules on Mantle (see §5.1). HyperTrade remains a non-custodial interface for wallet keys and trading; it does not hold user funds.

| | |
|---|---|
| **User Experience** | Expo app, onboarding, portfolio, charting, quick trade, IBAN accounts, cards roadmap |
| **Wallet Layer** | Privy embedded EOA, ERC-4337 smart wallet, hidden Solana account, user-side signing |
| **UR.APP** | External Wallet Access, URID, IBAN, fiat tokens, FX, cards |
| **Trading Layer** | Hyperliquid info/exchange APIs, API wallet setup, builder fee approval, WebSocket state |
| **Relay Layer** | Arbitrum USDC, EIP-7702 + Ambire batches, EIP-2612 permits, relayer pools |
| **Coordination Layer** | Railway replicas, Supabase locks, idempotency, worker leadership, notifications |

### System flow: from user intent to execution

| Mobile app | → | Privy wallet | → | Backend relay | → | Arbitrum + HL |
|---|---|---|---|---|---|---|
| User chooses market, size, leverage, and transfer intent. | | User signs permits or EIP-712 authorizations on device. | | Validates auth, assigns relayer, coordinates locks and retries. | | USDC settles on Arbitrum; trading executes on Hyperliquid. |

# 5. Wallet, Custody, and Gasless Infrastructure

HyperTrade uses Privy to make wallet ownership accessible. Users can authenticate through email, Google, or Apple. The app creates an embedded Ethereum wallet for users without one, wires a Privy smart wallet for ERC-4337 account abstraction, and defensively creates a Solana wallet that remains hidden from the UI today. Arbitrum is the active EVM chain.

**User-controlled**

- Privy wallet keys
- EIP-712 approvals
- 7702 + Ambire batches
- USDC / fiat permits

**Backend-controlled**

- Relayer gas payment
- Nonce locks
- Market data orchestration
- Operational records

TRUST BOUNDARY

## Gasless relayer design

Users sign EIP-2612 USDC permits on device. The backend cannot move funds by itself; it can only submit the exact action authorized by the user. A deterministic relayer assignment maps each wallet to one relayer, and Supabase locks serialize nonce-sensitive execution across Railway replicas.

### Gasless USDC permit relay

**Permit** — User signs EIP-2612 approval for a scoped spender and amount.

→ **Assign** — Backend hashes the user address to select a relayer.

→ **Lock** — Supabase prevents concurrent nonce collisions for that relayer.

→ **Execute** — Relayer pays gas and submits permit + transferFrom.

## 5.1 UR.APP Integration: External Wallet Access and IBAN Accounts

**UR.APP** is the regulated banking infrastructure that powers HyperTrade's IBAN accounts, fiat-token balances, FX, pay-in/payout, and debit card rails. HyperTrade is a partner interface on top of UR.APP's APIs and onchain contracts — not a custodian. Developer documentation lives at [docs.ur.app](docs.ur.app); contract addresses, ABIs, and audit reports are in the [smart contracts](smart contracts) reference.

## How HyperTrade integrates with UR.APP

UR.APP integrations are built from three independent choices: **Account Mode** (who holds fiat and who signs), **Card Mode** (what funds card spend), and **KYC Mode** (how identity is verified). HyperTrade's open-source reference implementation enables UR.APP by default, but **builders forking the codebase can omit UR.APP entirely** and ship trading-only flows — see §5.1 custody boundary.

| UR.APP integration choice | What UR.APP offers | What HyperTrade uses |
| --- | --- | --- |

| | | |
|---|---|---|
| Account Mode | Managed Custody (partner signs fiat) · External Wallet Access (user wallet signs) | External Wallet Access — user signs; UR rules apply to fiat tokens |
| Card Mode | Fiat Only (spend fiat balance) · Crypto Backed (real-time auth against crypto) | Fiat Only — fiat-backed card (roadmap) |
| KYC Mode | UR.APP webview · Sumsub SDK · Sumsub reuse | Sumsub SDK in the HyperTrade mobile app |

**External Wallet Access (HyperTrade's Account Mode):** the user's Privy embedded EOA owns the onchain **URID** (ERC-721 identity NFT) and receives **fiat-token balances** (e.g. USD24, EUR24) at the user's wallet address on Mantle. Banking actions require the user to sign — via EIP-712, EIP-2612 permits, or EIP-7702 batched execution — while HyperTrade sponsors gas where possible. This is not the same as holding raw USDC: once USDC is deposited into UR.APP contracts, credited balances are UR.APP-regulated fiat tokens subject to UR.APP's terms and compliance controls (see custody boundary below).

**Managed Custody (not used by HyperTrade):** in UR.APP's alternative Account Mode, fiat lives in a UR.APP-managed account and the partner backend signs fiat actions on the user's behalf. This suits consumer neobanks that hide wallet mechanics. HyperTrade deliberately chose External Wallet Access so users sign each banking action and URID ownership stays visible onchain at the user's EOA.

**Fiat Only cards (HyperTrade's Card Mode):** when card spend is enabled, the debit card draws from the user's UR.APP fiat balance (tokenized deposits). Users who want to spend crypto first off-ramp into fiat, then tap the card. **Crypto Backed** — UR.APP's alternative Card Mode where swipes settle against crypto via a partner prefund pool and real-time authorization webhooks — is not part of HyperTrade's design.

- **Design principle: UR.APP supplies regulated IBAN rails and onchain fiat tokens; HyperTrade supplies UX, Privy wallet connectivity, and gasless relay infrastructure — without becoming a custodian of wallet keys or trading balances.**

## Custody boundary: wallet USDC vs UR fiat tokens

HyperTrade's non-custodial posture applies to the core trading stack: Privy wallet keys stay on the user's device, Arbitrum USDC in the wallet before any deposit is user-controlled, and Hyperliquid trading balances move only under user-authorized API-wallet signatures. HyperTrade does not take custody of those assets.

When a user **opts in** to UR.APP banking and runs Add Money, USDC on Arbitrum is approved and deposited into UR.APP's onchain contracts (e.g. Fiat24CryptoDeposit). Settlement credits regulated fiat tokens such as **USD24** or **EUR24** on Mantle. In External Wallet Access mode those tokens appear at the user's EOA address, but they are **UR.APP-issued fiat tokens** moving through UR.APP's contract system — not generic self-custodied USDC. IBAN pay-in, FX, payout, and card spend are governed by UR.APP's smart-contract permissions, API policies, and licensed compliance program.

That regulated layer is expected to carry restrictions that do not apply to raw wallet USDC: KYC-gated account status, sanctions screening, AML monitoring, and — where law or UR.APP's compliance program requires — **freezes, holds, or blocks** on token movement. Users can typically off-ramp back to USDC in their own wallet via signed withdraw flows, but while balances remain in USD24/EUR24 form, **UR.APP's terms of service and regulatory obligations apply**; HyperTrade cannot override them.

- **Open-source builders: UR.APP IBAN, fiat-token, and card integration is optional. The reference HyperTrade app enables it, but a fork can ship Privy + Arbitrum USDC + Hyperliquid trading only — without UR.APP API credentials, Mantle banking contracts, or KYC flows.**

## Onchain identity and IBAN account model

- **URID (Fiat24Account):** each user receives an ERC-721 identity NFT minted to their Privy EOA on Mantle, gating access to IBAN accounts and card products. KYC status is recorded onchain in the contract's *status* mapping (e.g. Tourist → Live).
- **Fiat tokens at the EOA:** USD24, EUR24, CHF24, and related balances are credited to the user's Mantle address in External Wallet Access mode, but they are UR.APP-regulated ERC-20 fiat tokens — subject to UR.APP contract rules, URID status, and compliance controls (including freezes or holds where required).
- **IBAN provisioning:** after KYC reaches *Live*, UR.APP provisions SEPA/SWIFT IBAN details via the Profile API; inbound wires credit the user's fiat-token balance once settlement completes.

## KYC and identity verification (Sumsub)

Identity verification for IBAN accounts is delivered through **Sumsub**, integrated via UR.APP's compliance stack using the **Sumsub SDK in your app** KYC mode. Users complete document capture, liveness checks, and identity verification inside the Sumsub mobile SDK surfaced from HyperTrade.

HyperTrade does **not** store user identity documents, biometric captures, or other sensitive KYC payloads on its own servers. The app embeds the Sumsub SDK for the capture UX and exchanges short-lived access tokens through UR.APP API endpoints (e.g. account-status and create-access-token flows). HyperTrade backends receive only the *outcome* needed to gate product features — verification status, chain eligibility, and compliance state — not copies of passports, selfies, or raw document files.

- **SDK on device:** document and liveness capture runs in Sumsub's SDK; sensitive media stays in Sumsub / UR.APP supervised infrastructure.
- **API orchestration only:** HyperTrade calls UR.APP endpoints to mint SDK tokens and poll verification status.
- **No doc vault:** HyperTrade operational databases hold wallet links, job state, and feature flags — not KYC document archives.
- **UR.APP-led retention:** document retention, AML obligations, and regulatory record-keeping are governed by UR.APP and Sumsub under their licenses.

## Gasless flows: EIP-7702, Ambire, and permit relays

Several UR.APP banking actions require onchain authorization. HyperTrade sponsors gas so users never need native tokens for Mantle or Arbitrum operations. Two relay patterns cover the majority of flows:

| Flow | Chain | Authorization pattern | What the user signs |
|---|---|---|---|
| Add Money (USDC → fiat) | Arbitrum → Mantle (LZ) | EIP-7702 + Ambire batched execute | 7702 SetCode authorization (once per chain) + Ambire batch signature for approve + deposit |
| FX Convert (fiat ↔ fiat) | Mantle | EIP-7702 + Ambire batched execute | Same 7702 delegation + batch signature for approve + moneyExchangeExactIn |
| Withdraw / Payout | Mantle | EIP-2612 permit (REST settlement) | EIP-2612 permit authorizing UR.APP to debit fiat tokens |
| Hyperliquid deposit | Arbitrum | EIP-2612 USDC permit | Scoped USDC permit for relayer transferFrom |

**EIP-7702 delegation:** on first use per chain, the user signs an EIP-7702 authorization that temporarily delegates their EOA to Ambire's audited AmbireAccount7702 implementation. The EOA's code slot becomes *0xef0100 + delegate address*, enabling batched execution with *_msgSender() == user EOA* so UR contracts (e.g. Fiat24CryptoDeposit) resolve the correct URID owner.

**Ambire batch execution:** the user signs an off-chain Ambire batch hash (raw secp256k1, no EIP-191 prefix). HyperTrade's dedicated UR.APP relayer pool broadcasts a type-4 (EIP-7702) transaction that attaches the authorization, invokes *EOA.execute(calls[], signature)*, and pays gas plus any LayerZero cross-chain fees. Relayer assignment is deterministic per wallet with Supabase locks to prevent nonce collisions across Railway replicas.

**Add Money: EIP-7702 gasless deposit**

**Quote**
User enters USDC amount; backend fetches UR.APP quote + contract addresses.

→

**Delegate**
User signs 7702 SetCode auth binding EOA to Ambire delegate (once per chain).

→

**Batch sign**
User signs approve + depositTokenViaUsdc calls via Ambire batch hash.

→

**Relay**
HyperTrade relayer broadcasts type-4 tx; LZ bridges USDC; UR contracts

## Operational verbs enabled today and on roadmap

- **Pay-in:** user wires fiat to their IBAN; balance credits on settlement.
- **Payout:** user signs and sends fiat from IBAN balance to an external beneficiary via SEPA/SWIFT.
- **On-ramp / Add Money:** USDC on Arbitrum deposits into UR.APP contracts; Mantle balance credits as USD24/EUR24 (regulated fiat tokens, not raw USDC).
- **Off-ramp / Withdraw:** fiat tokens convert back to USDC in the user's wallet via permit-authorized settlement.
- **FX:** gasless fiat-to-fiat conversion on Mantle via 7702 + Ambire.
- **Debit card (roadmap):** Fiat Only mode — card spend draws from UR.APP fiat balance; card credentials revealed in UR.APP secure webview.

Withdraw and payout flows use UR.APP's permit REST API: the user signs an EIP-2612 permit on device, and UR.APP submits and pays for onchain settlement — HyperTrade's relayer does not move funds the user did not authorize.

# 6. Hyperliquid Execution, Non-Crypto Perpetuals, and Outcome Markets

Hyperliquid is the active execution venue for HyperTrade. The app integrates Hyperliquid info APIs, exchange APIs, WebSocket state, API-wallet order signing, builder fee approvals, and environment-scoped caches for clean separation between live and test environments.

One of the most important market shifts is that onchain perpetual markets are expanding beyond crypto. Public reporting in early 2026 cited Hyperliquid HIP-3 markets reaching roughly $1.4B in open interest, with oil and other non-crypto markets driving significant demand. This is aligned with HyperTrade's broader thesis: users do not only want memecoins and low-utility tokens; they want access to global macro, commodities, FX, indices, and equities in one mobile interface.

HyperTrade can tap HIP-3 providers such as XYZ Markets and other deployers as they bring more non-crypto markets onchain. This gives the app a path to diversify away from bloated crypto-only listings and toward markets users already understand: oil, gold, forex, large equities, and major indices.

A second category worth watching is outcome trading. Hyperliquid's HIP-4 work is publicly described by ecosystem coverage as outcome or prediction-market infrastructure, live on testnet and aimed at binary/range outcomes that can resemble prediction markets or bounded options. Prediction markets such as Polymarket and Kalshi have shown strong consumer demand for event-based trading; if HIP-4 matures on mainnet, HyperTrade can extend the same mobile interface principles to outcomes without rebuilding the underlying market infrastructure.

For HyperTrade, this matters strategically: perpetuals, non-crypto HIP-3 markets, and future outcome markets all point in the same direction — onchain markets are becoming a general interface for trading views about the world, not only crypto token prices.

**Hyperliquid trading setup**

**Fund**
User moves USDC into trading balance.

→

**Approve**
User approves API wallet and builder fee cap.

→

**Sign**
API wallet signs fast order flow.

→

**Execute**
Hyperliquid validates margin, fees, and market rules.

# 7. Revenue Model and Sustainable Growth

HyperTrade is designed around a lightweight, scalable revenue model. The product does not need to operate a matching engine, custodian, market maker balance sheet, or exchange back office to begin monetizing. Instead, it sits as a high-quality interface layer on top of mature external infrastructure: Privy for wallet onboarding, Arbitrum for USDC rails, UR.APP for IBAN accounts, Supabase for operational state, Railway for backend deployment, and Hyperliquid for execution.

The current revenue model is the Hyperliquid builder model. Users approve a transparent builder fee cap during one-tap setup, and orders can include HyperTrade's builder code. This lets the app monetize usage while keeping execution on Hyperliquid and custody with the user. The key alignment is simple: revenue grows with real trading activity and product adoption, not with custody, hidden spreads, or user deposits sitting on a company balance sheet.

| Revenue Stream | Status | Why It Scales |
|---|---|---|
| Hyperliquid builder fees | Active infrastructure | Usage-based revenue from order flow through the interface, with user-approved fee caps. |
| Cards and payments | Roadmap | Potential partner revenue from card activity, IBAN top-ups, FX/spend programs, or negotiated revenue share with UR.APP. |

| | | |
|---|---|---|
| AI-assisted agents | Active infrastructure | House LLM + market-data pipeline on the worker; monetization aligned with builder fees on agent-originated flow. |

## Cost structure

- **Low fixed cost:** Railway, Supabase, Privy, RPC providers, notifications, analytics, and standard monitoring are manageable compared with operating exchange or custody infrastructure.
- **Usage-aligned upside:** builder revenue increases as real trading usage increases.
- **Operational leverage:** more users do not require a proportional increase in headcount or servers because core coordination is stateless and lock-driven.
- **Runway discipline:** modest infrastructure costs reduce pressure to over-monetize early users or rush token launches before the regulatory environment is ready.

**Business model principle: monetize the interface layer transparently while keeping custody, settlement, and market execution in specialized infrastructure.**

# 8. AI Agents and Market Intelligence

HyperTrade combines two AI surfaces: **in-app market analysis** (Gemini-backed research for human traders) and **AI Trading Agents** — autonomous copilots that monitor Hyperliquid perps, reason over structured market context, and place orders through HL named-agent keys the user explicitly approves.

The agent stack is split into a **control plane** and an **execution plane**. The mobile app and FastAPI backend handle agent CRUD, Privy JWT auth, wallet-ownership checks, and the approve-agent ceremony. A dedicated **ai-agent-worker** service (Node/TypeScript on Railway) runs the trading brain and HL order flow — it holds no public HTTP surface and talks to Supabase with the service role only.

### AI agent runtime (hourly cycle)

**Ingest**
Per-symbol market cache: futures OHLC, OI, funding, flows, ETF demand, macro

→

**Reason**
Opening + winning/losing monitor prompts → structured JSON via user-chosen LLM.

→

**Guard**
Schema validation, budget/leverage caps, symbol-conflict checks.

→

**Execute**
HL agent key signs orders/TP-SL; builder fee attached; decisions logged to

## Copilot mode and custody boundary

V1 ships **Copilot (shared wallet)** mode: agents trade from the user's main unified Hyperliquid balance, bounded by per-agent **max_capital_usd**, optional per-position caps, leverage limits, and a symbol allowlist (up to 20 coins). Each agent receives its own HL keypair; the user approves it as a **named agent** (*approveAgent*) alongside the device's manual trading agent. HL protocol guarantees agent keys can sign L1 order actions only — not withdrawals — even if backend data were tampered with.

Agent private keys are generated in the worker, encrypted with **AES-256-GCM** under a Railway-only *AGENT_KMS_KEY*, and stored as ciphertext in Supabase. The backend never holds LLM API keys; model inference runs on the worker using house provider keys (Gemini, Grok, OpenAI, DeepSeek, Claude). Users can pause, stop, or revoke agents at any time.

## Brain: data → flags → prompts → validation

Market context is assembled in two phases each cycle. **Phase 1** builds a shared per-symbol snapshot once per coin — aggregated futures and spot history, open interest, funding, taker flow, liquidations, premium/basis vs spot, plus symbol-specific overlays: Deribit DVOL options context for BTC/ETH, spot-ETF flow history for BTC/ETH/SOL/XRP, and a macro calendar slice (US holidays + high-impact CPI/FOMC windows). Slow-moving context is persisted in a **global_context_cache** table so all agents and worker restarts share one refresh, not N duplicate pulls.

**Phase 2** fans out per active agent. Raw series feed **computeScalperFlags** and a composite long/short score. The brain then branches:

- **Opening path** — for flat symbols: opening prompt with session context, conviction bands, planned stop anchors, and macro/ETF sections. Responses are JSON-validated; sub-conviction setups are skipped until market context updates.
- **Winning monitor** — for profitable positions: hold / add / trim / exit with stop-management rules (breakeven, tighter) that never loosen protective stops.
- **Losing monitor** — for underwater positions: hold / cut / exit with stored opening thesis and invalidation criteria replayed into the prompt.

LLM output is treated as **untrusted input**. Validators clamp leverage, size, symbol allowlists, and reduce-only integrity before the HL adapter sees an order. Live mids from Hyperliquid (cycle-cached) anchor decisions; cached OHLC snapshots are fallback when mids are unavailable. Opening decisions for the same symbol + model dedupe within a cycle so parallel agents do not burn duplicate LLM calls.

## Execution adapter and conflict guards

The HL execution adapter is a server-side port of HyperTrade's signing semantics: unified-account free margin, position-linked TP/SL via *positionTpsl*, reduce-only IOC exits with in-cycle retries, and HyperTrade's builder fee on every agent order. Budget headroom for new opens uses live mark notional so inflated winners consume cap honestly.

| Guard | Behavior |
|---|---|
| User conflict | Skip opening a coin where the user already holds a manual position the agent does not own. |
| Peer conflict | Copilots on the same master wallet run sequentially and claim symbols so two agents cannot race the same coin. |
| Manual close | If the user closes or reduces an agent position externally, the worker adopts reality and does not re-open. |
| Order tagging | Agent orders carry a deterministic *cloid* prefix so portfolio UI can badge bot-managed coins. |

Every cycle writes an audit trail: **ai_agent_decisions** (structured decision + reasoning JSON), **ai_agent_positions** (thesis, stops, PnL state), and **ai_agent_runs** (equity snapshot for dashboard charts). Supabase RLS is deny-all on agent tables — only the backend and worker service role can read or mutate rows, scoped by *privy_user_id* at the API layer.

**Product principle: AI agents automate research and execution within explicit user-approved bounds — named HL agent keys, not custodial trading accounts.**

# 9. Community, Token Philosophy, and Roadmap

HyperTrade's FAQ currently states a conservative position on a token or airdrop: early users should be rewarded, but token distribution is postponed until the regulatory landscape is clearer. The planned philosophy is user ownership, not rushed speculation.

The stated direction is a future HTX distribution with a large majority allocation to users, tied to real usage, loyalty, trading activity, referrals, and feature engagement. This framing matters: the app should reward authentic participation rather than passive farming.

## Roadmap themes

- Improve the core Hyperliquid trading interface for both casual and advanced users.
- Expand stablecoin utility through UR.APP IBAN accounts, Fiat Only card partnerships, and everyday finance use cases.
- Monitor Hyperliquid HIP-4 outcome markets as a potential future surface for prediction-market style user demand.
- Operate AI Trading Agents (copilot mode) with worker-side brain, global market cache, and HL named-agent signing.
- Maintain non-custodial architecture as the product expands.
- Stay deliberate on token design until compliance and market structure are clearer.

# 10. Risk, Security, and Trust Model

HyperTrade's security model is narrow by design. User wallets sign. Relayers relay. Hyperliquid executes. Supabase coordinates operational state. This separation reduces the blast radius of any one component and keeps wallet-key custody with users for the trading stack; optional UR.APP fiat-token flows add a regulated compliance layer governed by UR.APP (§5.1).

| Risk | Current Control |
|---|---|
| User key custody | Privy device-side embedded wallet model; backend never stores user private keys. |
| Gasless relay abuse | User-signed permits, deterministic relayer assignment, replay checks, rate limits. |
| Replica nonce collision | Supabase relayer locks with TTL recovery. |
| Setup incompleteness | Setup considered complete only when agent is active and builder fee is approved. |
| Third-party dependency | Clear dependency on Privy, Arbitrum, Mantle, Hyperliquid, UR.APP, oracles, and card partners. |

| | |
|---|---|
| 7702 relay abuse | User-signed 7702 authorizations and Ambire batches; relayer bound to URID owner; rate limits and inflight deposit guards. |
| Regulatory uncertainty | Non-custodial interface posture; UR.APP-led KYC/AML for regulated IBAN and card rails. |
| UR fiat-token compliance | USD24/EUR24 balances use UR.APP contracts; lawful freezes, holds, and KYC-gated restrictions are expected on the regulated rail — not applicable to raw wallet USDC or Hyperliquid-only usage. |
| AI agent key custody | Per-agent HL keys encrypted AES-256-GCM; user approves named agent; HL blocks withdrawals from agent keys. |
| LLM output abuse | Structured JSON validators + adapter hard caps on size, leverage, symbols; untrusted model output never sent raw to HL. |
| KYC data handling | Sumsub SDK on device; HyperTrade stores verification outcomes only — no identity documents or biometric archives on HyperTrade servers. |

# 11. Regulatory and Policy Landscape

HyperTrade should be understood as a user interface and wallet-connected application, not as a custodian. This distinction is increasingly important as regulators clarify how self-custodial trading interfaces fit within existing frameworks.

In April 2026, SEC Division of Trading and Markets staff issued a statement on broker-dealer registration for certain user interfaces used to prepare crypto asset securities transactions. The statement is not a Commission rule and has no independent legal force, but it is directionally important: it describes conditions under which staff would not object to certain covered user interface providers operating without broker-dealer registration, including self-custodial wallet contexts, objective parameters, educational material, disclosures, and no custody of user funds.

HyperTrade's design philosophy is consistent with several principles highlighted in that staff statement: users control wallets, the app prepares transaction parameters, fees and limitations should be disclosed, and the interface should avoid discretionary control over user funds or execution decisions. This is not legal advice, and product rollout must continue to be reviewed jurisdiction by jurisdiction.

Separately, the Hyperliquid Policy Center launched in February 2026 as an independent research and advocacy organization focused on decentralized market infrastructure, perpetual derivatives, and practical regulatory frameworks. Its existence is a positive ecosystem development for interfaces like HyperTrade because policy clarity around decentralized markets can reduce uncertainty for builders, partners, and users.

# 12. Appendix: Factual Anchors, Audits, and Sources

## Regulatory and ecosystem references

- SEC Division of Trading and Markets staff statement, April 13 2026: covered user interfaces, self-custodial wallet contexts, objective parameters, disclosures, and no custody of user funds. The statement is staff view only and not a Commission rule.

- Hyperliquid Policy Center launch announcement, February 18 2026: independent 501(c)(4) research and advocacy organization focused on decentralized finance, perpetual derivatives, and blockchain-based financial infrastructure.
- Hyperliquid Policy Center About page: describes Hyperliquid as a public, permissionless blockchain and decentralized exchange known for perpetuals, transparency, accessibility, performance, and non-custodial use.
- Public market reporting in 2026 cited HIP-3 open interest around the $1.4B range and highlighted oil / non-crypto perpetuals as major drivers of growth.

## UR.APP developer documentation

- **Developer docs home:** docs.ur.app
- **Integration guide:** Account Mode, Card Mode, and KYC Mode choices — integration guide
- **External Wallet Access:** HyperTrade's Account Mode — External Wallet Access docs
- **Managed Custody:** alternative Account Mode (not used by HyperTrade) — Managed Custody docs
- **Smart contracts:** Fiat24Account, fiat tokens, deposit/relay ABIs on Mantle — smart contracts
- **Contract source & audits:** ur-app/ur-contracts · audit reports

## Third-party security audits

HyperTrade relies on audited infrastructure from wallet, execution, relay, and UR.APP partners. Independent audit reports for key dependencies:

- **Ambire EIP-7702 delegate (Hunter Security):** audit of AmbireAccount7702 used for gasless batched execution — view audit report
- **UR.APP onchain contracts (Blocksec):** Fiat24 deposit, relay, on-ramp, card authorization, and Mantle contract suites — view audits on GitHub
- **LayerZero (Endpoint V2, OFT, index):** cross-chain messaging and omnichain token bridges for Add Money (Arbitrum → Mantle) — Endpoint V2, OFT, full index
- **Arbitrum native USDC (Trail of Bits):** audit of Arbitrum USDC custom gateway and ArbOS upgrade — view audit report
- **Hyperliquid (Zellic):** audit of Hyperliquid core infrastructure — view audit report
- **Privy Shamir Secret Sharing (Zellic):** audit of Privy embedded-wallet key-sharding architecture — view audit report

**HyperTrade codebase anchors:** Privy embedded EOA + ERC-4337 smart wallet; External Wallet Access (URID, IBAN, balances, history); gasless EIP-7702 + Ambire Add Money and FX on Mantle; EIP-2612 permit relays for Hyperliquid deposits and UR.APP withdraw/payout; dedicated UR.APP relayer pool with Supabase nonce locks; Hyperliquid API-wallet + builder-fee setup; frontend margin and liquidation guards; ai-agent-worker (leader-gated cycles, global market cache, multi-LLM brain, HL named-agent adapter); Sumsub KYC via mobile SDK with verification outcomes only — no document storage on HyperTrade servers.